

END TERM EXAMINATION

FIFTH SEMESTER [B.TECH] JANUARY 2024

Paper Code: CIC-309

Subject: Software Engineering

Time: 3 Hours

Maximum Marks: 75

Note: Attempt five questions in all including Q. No.1 which is compulsory. Select one question from each unit. Use of simple Calculator is allowed.

- Q1 Answer **any five** questions:- (5x3=15)
- Briefly explain the concept of Software Crisis and how it differs from common Software Myths. Provide real-world examples.
 - Distinguish software re-engineering and reverse engineering?
 - Calculate and interpret Halstead's effort for a program where the number of distinct operators is $n_1=15$, the number of distinct operands is $n_2=20$, the total number of operators is $N_1=120$, and the total number of operands is $N_2=80$.
 - Discuss the implications of high coupling on software maintainability, with examples from industry practices.
 - Discuss the role of regression testing in Agile methodologies.
 - Draw the Context level 0 and level 1 DFD for the Safe home Software?
 - Assess the effectiveness of data dictionaries in managing large-scale software projects, citing real-world applications.
 - Develop a set of test cases to perform boundary value analysis for the following input fields in an online registration form:
 - Age: The form requires students to enter their age, which must be between 18 and 30 years inclusive.
 - Number of Completed High School Subjects: The form also asks for the number of completed high school subjects, which should be a minimum of 5 and a maximum of 12.
- Ensure your test cases cover the boundary conditions for these input fields.

UNIT-I

- Q2 (a) Compare and contrast the Waterfall and evolutionary Models of SDLC. Focus on key stages, flexibility, risk management, and suitability for different types of projects. Provide examples. (7.5)
- (b) Explain the importance and steps of requirement elicitation in software engineering. Discuss the FAST and Use Case approaches, highlighting their strengths and limitations. (7.5)
- Q3 (a) Describe the key characteristics and structure of an effective Software Requirement Specification (SRS) document. Use IEEE standards as a reference. (7.5)
- (b) Discuss the differences and similarities between Functional and Non-Functional requirements with examples. How do these impact the software development process? (7.5)

P.T.O.

[-2-]

UNIT-II

- Q4 (a) Explain the COCOMO cost estimation model. Perform a basic COCOMO calculation for a project with a given size (50 KLOC). Using the Intermediate COCOMO model, calculate the effort in person-months. Assume the project has 'HIGH' complexity [CMLX=High=1.15]. (5)
- (b) Describe the concepts of cohesion and coupling in software design. Provide examples of each and discuss their impact on software maintainability and flexibility. (5)
- (c) Write a short note on live variables and module weakness. (5)
- Q5 (a) Discuss the Putnam Resource Allocation Model, including its application and limitations. Perform a calculation using given project parameters. (5)
- (b) Compare and contrast Object-Oriented Design (OOD) and Function-Oriented Design (FOD) in terms of their methodology, advantages, and disadvantages. (5)
- (c) Differentiate between Function Point and LOC software Metrics. Why Function points metrics are preferred over Lines of Code Metrics. (5)

UNIT-III

- Q6 (a) Explain Halstead's Software Science metrics. Write equations to calculate the Effort and Time for a given program using Halstead's methods. (7.5)
- (b) Describe the concept of Software Reliability. Compare the Basic Model and Logarithmic Poisson Model in terms of their approach and applicability. (7.5)
- Q7 (a) Discuss the importance of software metrics in software engineering. Explain Token Count and Information Flow metrics with examples. (7.5)
- (b) Explain the CMM (Capability Maturity Model) and ISO 9001 standards. Discuss how they contribute to software quality assurance. Give example of three CMM Level 3 companies based in India. (7.5)

UNIT-IV

- Q8 (a) Describe the testing process in software engineering, focusing on the importance of functional and structural testing. Provide examples of test cases for each. (5)
- (b) Create Control Flow Graph for a module that checks if sum of 2 numbers is positive, negative or neutral. Also find cyclomatic complexity number for this module and number of independent paths in the module. (5)
- (c) Explain the concept and process of Software Maintenance. Discuss the Quick-fix and Boehm's models of maintenance with their merits and demerits. (5)
- Q9 (a) Define and differentiate between Verification and Validation in software testing. Provide examples to illustrate your points. (5)
- (b) Discuss the importance and methods of Software Configuration Management in the context of software maintenance. (5)
- (c) Explain the importance of acceptance testing in software development. Compare and contrast alpha and beta testing. (5)

CIC-309
p-1/2CIC-309
p-2/2